Streamlining IT Procurement: Automating Standard Laptop Orders with WI

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1. Abstract

This project focuses on making the process of ordering standard laptops in an organization faster and more reliable. At present, many companies still depend on emails, paper forms and manual checks for IT procurement. This creates delays, mistakes in budget tracking and confusion about who approved what. Our solution introduces an automated workflow system called WI (Workflow Intelligence) that handles procurement requests, manager approvals, budget verification, vendor selection and inventory updates in one connected flow. The system reduces human effort, keeps clear records and helps the IT and finance teams work together smoothly. The project shows how a simple web-based automation can turn a slow manual process into a quick and transparent one.

2. Introduction

Every company needs laptops for new employees, replacements and upgrades. Ordering these machines looks simple, but in real life it involves many people and many steps. An employee raises a request, the manager checks it, the finance team verifies the budget, the procurement team selects a vendor, and finally the inventory is updated. When these steps are done through emails and spreadsheets, things get lost or delayed.

This project aims to solve that problem by building an automated system for standard laptop orders. We call the system WI – Workflow Intelligence. It connects all the important stages of procurement into one digital path. The goal is not to replace people, but to remove repetitive work and make the process clear for everyone involved.

3. Problem Statement

In most organizations the current laptop procurement process has several pain points:

- Requests are sent by email or filled on paper forms that are hard to track.
- Manager approval often sits in inboxes for days.
- Budget checks are done late, sometimes after the order is already placed.
- Vendor selection depends on individual knowledge instead of clear rules.
- Inventory records are updated only after the laptop arrives, creating mismatch.
- There is no single place to see the status of any request.

These issues lead to longer waiting time for employees, higher chance of overspending and extra work for the IT and finance teams. A structured automated workflow is needed to fix these gaps.

4. Objectives

The main objectives of this project are:

1. To design a digital procurement request form for standard laptop models.
2. To build an automated approval flow that sends the request to the right manager.
3. To add an automatic budget check against the department's available funds.
4. To create simple rules for vendor selection based on price, delivery time and preferred partners.
5. To update inventory records as soon as an order is confirmed.
6. To provide a clear dashboard so that requestors, managers and procurement staff can see status in real time.
7. To keep a complete audit trail of every decision taken during the process.

5. Existing System

In the present setup, an employee who needs a laptop usually writes an email to the IT helpdesk or fills a shared Excel sheet. The helpdesk then forwards the request to the reporting manager for approval. After the manager replies, the request goes to the finance team for budget confirmation. Only then does the procurement team look for a vendor, compare quotations and place the order. Once the laptop is delivered, someone from IT updates the asset register manually.

This chain depends heavily on people remembering to forward emails and update files. There is no automatic reminder, no single source of truth and no easy way to measure how long each step takes. As a result, a simple laptop order can take two to three weeks even when the machine is available in stock or from a preferred vendor.

6. Proposed System

The proposed system, WI (Workflow Intelligence), is a web-based application that manages the entire standard laptop order process from request to delivery confirmation. All standard laptop models and their approved configurations are stored in a catalogue. When a user raises a request, the system already knows the approximate cost and preferred vendors.

Key features of the proposed system include:

- Role-based access for employees, managers, finance officers and procurement staff.
- Automatic routing of the request to the correct manager based on the organisation chart.
- Instant budget validation against the department's remaining allocation.
- Pre-defined vendor ranking rules that suggest the best supplier.
- Real-time status tracking and email or chat notifications at every stage.
- Automatic creation of a purchase order and update of the inventory module once the order is confirmed.

The system is designed to work with existing company tools such as email and chat platforms, so users do not need to learn a completely new way of working.

7. System Architecture & Workflow

The system follows a simple layered architecture: a user interface layer, a workflow engine, a rules engine for budget and vendor decisions, and a data layer that stores requests, approvals, purchase orders and inventory records.

Below is the step-by-step workflow for a standard laptop order:

Step 1 – Raise Procurement Request

The employee logs into WI and selects a standard laptop model from the approved catalogue. He or she enters the reason for the request (new joiner, replacement, upgrade) and the required delivery date. The system creates a unique request ID and shows the estimated cost.

Step 2 – Manager Approval

The request is automatically sent to the employee's reporting manager. The manager receives a notification and can approve, reject or ask for more information inside the same system. A reminder is sent after 48 hours if no action is taken.

Step 3 – Budget Check

Once the manager approves, the workflow engine checks the department's remaining budget for IT assets. If sufficient funds are available, the request moves forward. If not, the system notifies both the manager and the finance team so they can either release additional budget or cancel the request.

Step 4 – Vendor Selection

The rules engine looks at the list of preferred vendors, current prices, average delivery time and past performance. It ranks the vendors and suggests the top choice. A procurement officer can accept the suggestion or choose another vendor with a short justification. This keeps the process fast while still allowing human control when needed.

Step 5 – Purchase Order & Inventory Update

After vendor confirmation, WI generates a purchase order and sends it to the selected vendor. At the same time, the inventory module records the expected arrival of the laptop against the request ID. When the laptop is received and checked by the IT team, the inventory status changes from "On Order" to "In Stock / Assigned".

Step 6 – Closure & Notification

The original requestor is informed that the laptop is ready for collection or has been delivered. All documents (request form, approvals, purchase order, delivery note) are stored against the request ID for future audits.

8. Benefits

Automating standard laptop orders with WI brings several clear benefits:

- Faster turnaround – most requests can be completed in a few days instead of weeks.
- Better budget control – no order moves forward without a successful budget check.
- Less manual work – employees and managers spend less time chasing emails.
- Transparent status – anyone involved can see exactly where a request is stuck.
- Consistent vendor decisions – selection follows agreed rules rather than personal preference.
- Clean audit trail – every approval and change is recorded automatically.
- Improved inventory accuracy – stock levels stay up to date without extra data entry.

9. Conclusion

This project shows that a well-designed automated workflow can remove many of the delays and errors found in traditional IT procurement. By connecting the procurement request, manager approval, budget check, vendor selection and inventory update into one system, WI makes the process faster, clearer and easier to control. The solution is especially useful for standard laptop orders that follow the same path every time. With simple technology and clear business rules, organisations can give employees the tools they need without creating extra work for support teams.

10. Future Scope

In the future the system can be extended in several useful ways:

- Support for other IT assets such as monitors, headsets and mobile phones.
- Integration with the company's ERP or accounting software for automatic invoice matching.
- Mobile application so managers can approve requests while travelling.
- Predictive analytics that forecast laptop demand based on hiring plans.
- AI-based recommendation of laptop models according to the user's job role.
- Direct API connection with major vendor portals for real-time price and stock checks.

11. References

1. Project Management Institute. A Guide to the Project Management Body of Knowledge (PMBOK Guide).

2. ISO 20400:2017 – Sustainable Procurement Guidance.

3. Business Process Model and Notation (BPMN) 2.0 Specification, Object Management Group.

4. Digital Transformation of Procurement – academic papers and industry white papers on workflow automation.

5. Internal company process documents related to IT asset management and purchase procedures (used for requirement study).